From Classroom/Integrated School Programs 7 in Top 20, 23 in Top 100, 54 in Top 300, 106 in Top 500 All India Ranks & 2314 Students from Classroom /Integrated School Programs & 3723 Students from All Programs have been Awarded a Rank in JEE (Advanced), 2013

ALL INDIA TEST SERIES

FIITJEE **JEE (Advanced), 2014**

CONCEPT RECAPITULATION TEST - IV

Paper 2

Time Allotted: 3 Hours **Maximum Marks: 180**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part has only one section: **Section-A**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 to 08)** contains 8 multiple choice questions which have one or more correct answer. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

Section-A (09 to 16) contains 4 paragraphs with each having 2 questions. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

(ii) **Section-A (17 – 20)** contains 4 Matching Lists Type questions: Each question has four statements in LIST I & 4 statements in LIST II. The codes for lists have choices (A), (B), (C), (D) out of which only one is correct. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

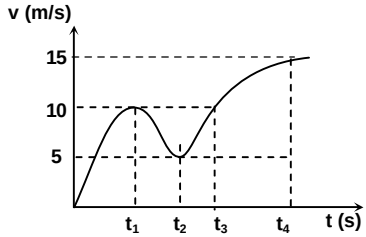
PART – I

SECTION – A

Single Correct Answer Type

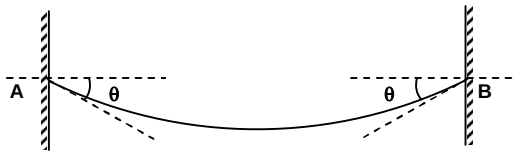
This section contain **8 multiple choice questions.** Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

1. Velocity time graph of a particle undergoing rectilinear motion is plotted as shown in the figure. Average acceleration of particle is zero between time intervals.



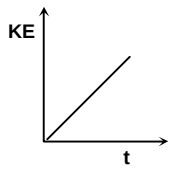
(A) 0 and t_1 (B) t_1 and t_2
(C) t_1 and t_3 (D) t_2 and t_4

2. A heavy string of mass m hangs between two fixed points A and B at the same level. The tangents to the string at A and B are at an angle θ with the horizontal as shown in the figure. The tension in the string at lowest point is



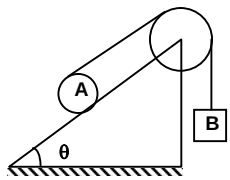
(A) $\frac{mg}{2\sin\theta}$ (B) $\frac{mg}{2\cos\theta}$
(C) $\frac{mg}{2\tan\theta}$ (D) $\frac{mg}{2\cot\theta}$

3. The kinetic energy of a body moving along a straight line varies with time as shown in the figure. The force acting on the body is



(A) zero
(B) constant
(C) directly proportional to velocity
(D) inversely proportional to velocity.

4. In the figure shown, a string is wound over a cylinder A. The other end of the string is attached to block B through a pulley. When the system is released the cylinder rolls down without slipping. The ratio of magnitude of vertical component of displacement of A and B in any time interval t is



(A) $\sin\theta : 1$ (B) $\sin\theta : 2$
(C) $\cos\theta : 1$ (D) $\cos\theta : 2$

Space for Rough work

5. Six stars of equal mass are moving about the centre of mass of the system such that they are always on the vertices of a regular hexagon of side length a. Their common time period will be

(A) $4\pi\sqrt{\frac{a^3}{Gm}}$ (B) $2\pi\sqrt{\frac{4\sqrt{3}a^3}{Gm\left(5\sqrt{3}+4\right)}}$

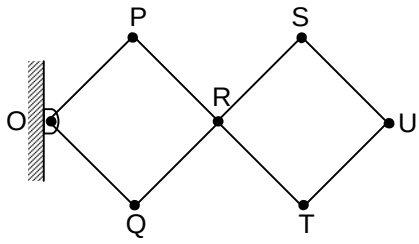
(C) $4\pi\sqrt{\frac{3a^3}{Gm}}$ (D) None of these

6. The end of a chain of length L and mass per unit length ρ, which is piled up on a horizontal platform is lifted vertically with a constant velocity u by a variable force F. Find F as a function of height x of the end above platform.

(A) $\rho\left(gx+2u^2\right)$ (B) $\rho\left(2gx+\rho u^2\right)$

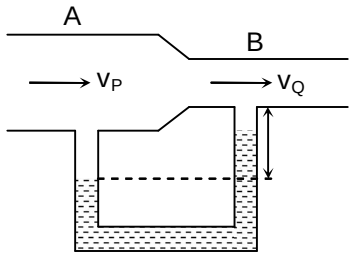
(C) $\rho\left(gx+u^2\right)$ (D) $\rho\left(u^2-gx\right)$

7. In the arrangement of rigid links of equal length l, they can freely rotate about the joined ends as shown in the figure. If the end U is pulled horizontally with constant speed 20 m/s, find the approx. speed of end P when the angle SUT is 90°.



(A) 5 m/s
(B) 10 m/s
(C) 7.1 m/s
(D) 14.12 m/s

8. Air is blown through a pipe AB at a rate of 1600 cm^3/sec. The cross-section area of the broad portion of pipe is 4 cm^2 and that of narrow portion is 2 cm^2. The difference in water length is (ρ_{air} = 1.4 kg/m^3, ρ_{water} = 10^3 kg/m^3)



(A) 1.6 mm (B) 3 mm

(C) 3.6 mm (D) None

Space for Rough work

Comprehension Type

This section contains **4 groups of questions.** Each group has 2 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct.

Paragraph for Questions 9 and 10

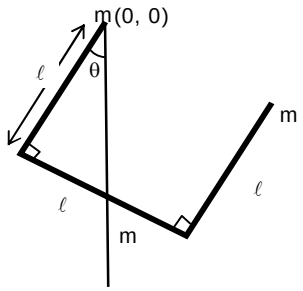
At certain pressure and 127^0 C temperature the mean kinetic energy of hydrogen molecules is 8×10^{-21} J. (Mass of hydrogen atom = 1.7×10^{-27} kg & atomic weight of nitrogen = 14). Then

9. Root mean square speed of hydrogen molecule at 27°C is
   (A) 0.88×10^4 m/sec (B) 1.88×10^4 m/sec.
   (C) 1.88×10^3 m/sec. (D) 2.88×10^3 m/sec

10. The rms speed of nitrogen molecules at 27^0C is
   (A) 5.02×10^2 m/sec. (B) 5.02×10^4 m/sec.
   (C) 2.51×10^2 m/sec. (D) 2.51×10^3 m/sec.

Paragraph for Questions 11 and 12

A 'U' shaped uniform rod of each of side having length 'ℓ' and mass 'm' is hinged from the one end of one of its limbs as shown.



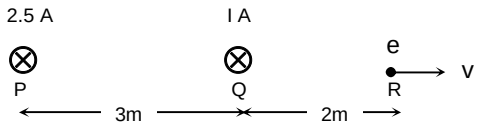
11. The Y_{cm} of the given system is
   (A) 2L/3 (B) –2L/3
   (C) –L/8 (D) –L/3

12. If the system is displace by small angle then time period of oscillation is
   (A) $\pi\sqrt{\frac{3\ell}{5g}}$ (B) $2\pi\sqrt{\frac{6\ell}{5g}}$
   (C) $\pi\sqrt{\frac{6\ell}{5g}}$ (D) $2\pi\sqrt{\frac{3\ell}{5g}}$

Space for Rough work

Paragraph for Questions 13 and 14

Two long parallel wires carrying currents 2.5 amp and I amp in the same direction as shown in figure. An electron moving with a velocity 4×10^5 m/sec. as shown in figure and experiences a force of magnitude 3.2×10^{-20} N at the point R. Then



13. The value of current I is
    (A) 2 amp.
    (B) 8 amp.
    (C) 4 amp.
    (D) 1 amp.

14. The position between PR at which a third long parallel wire carrying current 2.5 amp may be placed such that magnetic induction at R is zero.
    (A) at 1 m from R and the direction of current is inward.
    (B) at 1 m from R and the direction of current is outward
    (C) at 1 m from P and the direction of current is inward.
    (D) at 1 m from P and the direction of current is outward

Paragraph for Questions 15 and 16

A tube of length ℓ and radius R carries a steady flow of a fluid whose density is ρ and viscosity η. The velocity v of flow is given by $v = v_0[1 - (r^2/R^2)]$, where r is the distance of flowing fluid from the axis. Then

15. Kinetic energy of the fluid within the volume of the tube is
    (A) $\frac{1}{6}\pi\ell R^2\rho v_0^2$
    (B) $\frac{1}{8}\pi\ell R^2\rho v_0^2$
    (C) $\frac{1}{4}\pi\ell R^2\rho v_0^2$
    (D) $\frac{1}{3}\pi\ell R^2\rho v_0^2$

16. The difference of pressure at the ends of the tube is
    (A) $\frac{4\eta\ell v_0}{R^2}$
    (B) $\frac{8\eta\ell v_0}{R^2}$
    (C) $\frac{2\eta\ell v_0}{R^2}$
    (D) $\frac{\eta\ell v_0}{R^2}$

Space for Rough work

Matching list Type

This section contains **4 multiple choice questions**. Each question has matching lists. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

17. A block of mass m is placed on a rough horizontal surface having coefficient of friction μ. The magnitude of external horizontal force applied on the block gradually increases. The force exerted by the block on the surface is N. Match **List - I** with **List - II** and select the correct answer using the codes given below the lists:

List – I		**List – II**	
(P)	The magnitude of N	(1)	is less than or equal to $\tan^{-1}(\mu)$.
(Q)	N is less than of equal to	(2)	decreases gradually.
(R)	The angle made by N with horizontal.	(3)	$\sqrt{(mg)^2+(\mu mg)^2}$
(S)	The angle made by N with vertical	(4)	increases gradually.

Codes:

	P	Q	R	S
(A)	2	3	1	4
(B)	1	4	2	3
(C)	4	3	2	1
(D)	2	3	4	1

18. Match **List - I** with **List - II** and select the correct answer using the codes given below the lists:

List – I		**List – II**	
(P)	For the propagation of mechanical Wave	(1)	have zero displacement simultaneously at some instant.
(Q)	In a stationary wave, all the particles between two nodes.	(2)	no material medium is required
(R)	In a transverse progressive wave, all the particles of the medium.	(3)	Material medium is required.
(S)	For the propagation of e.m. wave	(4)	have the same time period and amplitude of vibration through out the wave.

Codes:

	P	Q	R	S
(A)	2	3	1	4
(B)	1	4	2	3
(C)	4	1	2	3
(D)	3	1	4	2

Space for Rough work

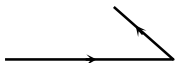
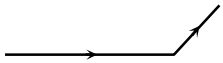
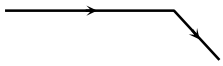
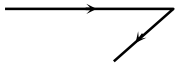
19. Match **List - I** with **List - II** and select the correct answer using the codes given below the lists:

List – I		**List – II**	
(P)	For a sustained interference pattern of light.	(1)	amplitude of waves from the two sources must be same.
(Q)	Newton's corpuscular theory	(2)	could not explain the phenomenon of interference of light.
(R)	Huygens wave theory	(3)	coherent source is required.
(S)	For maximum contrast in the sustained interference pattern due to a YDSE.	(4)	could not explain photoelectric effect.

Codes:

	P	Q	R	S
(A)	2	3	1	4
(B)	1	4	2	3
(C)	3	2	4	1
(D)	2	3	4	1

20. Four rays of light parallel to optic axis and their path after passing through an optical system are shown in the following two columns. Match **List - I** with **List - II** and select the correct answer using the codes given below the lists:

List – I		**List – II**	
(P)		(1)	Convex lens
(Q)		(2)	Concave lens
(R)		(3)	Convex mirror
(S)		(4)	Concave mirror

Codes:

	P	Q	R	S
(A)	2	3	1	4
(B)	3	2	1	2
(C)	4	1	2	3
(D)	2	3	4	1

Space for Rough work

Chemistry

PART – II

SECTION – A

Straight Objective Type

This section contains **8 multiple choice questions.** Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE** are correct.

1. Which of the following is correct for critical temperature?
   (A) It is highest temp. at which liquid and vapour can coexist.
   (B) Beyond the critical temp. there is no distinction between two phases and gas can not be liquiefied by compression.
   (C) At critical temp., the surface tension of the system is zero.
   (D) At critical temp. the gas and the liquid phases have the different critical densities.

2. The correct expression(s) for an adiabatic process is/are:
   (A) $\frac{T_2}{T_1}=\left(\frac{V_1}{V_2}\right)^{\gamma-1}$ (B) $\frac{P_2}{P_1}=\left(\frac{T_1}{T_2}\right)^{\frac{\gamma-1}{\gamma}}$
   (C) $P_1V_1^{\gamma}=P_2V_2^{\gamma}$ (D) $P_1V_1^{\gamma-1}=P_2V_2^{\gamma-1}$

3. A liquid is in equilibrium with its vapour at its boiling point. On an average the molecules in the two phases have equal.
   (A) Intermolecular forces (B) Potential energy
   (C) Kinetic energy (D) Both kinetic energy and potential energy.

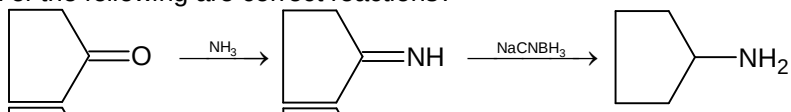
4. Choose the correct statement(s):
   (A) $BeCO_3$ is kept in the atmosphere of CO_2 since it is last thermally stable.
   (B) Be dissolves in an alkali solution forming $[Be(OH)_4]^{-2}$.
   (C) BeF_2 forms complex ion with NaF in which Be goes with cation.
   (D) BeF_2 forms complex ion with NaF in which Be goes with anion.

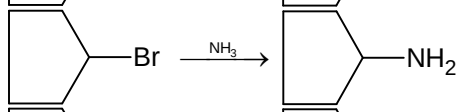
Rough Work

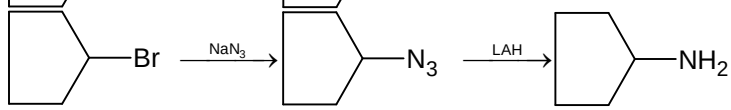
5. NH_3 can be obtained by
   (A) Heating of NH_4NO_3 or NH_4NO_2
   (B) Heating of NH_4Cl or $(NH_4)_2CO_3$
   (C) Heating of NH_4NO_3 with NaOH
   (D) Reaction of AlN or Mg_3N_2 or $CaCN_2$ with H_2O

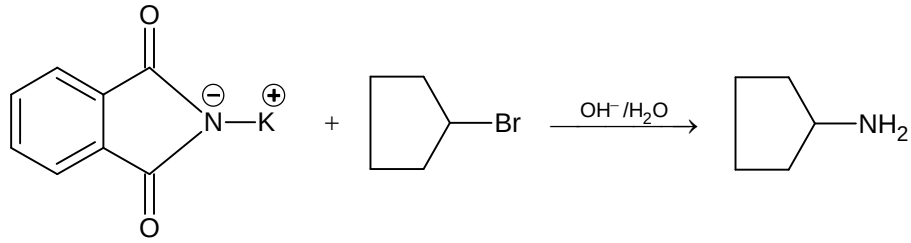
6. Among the following, the ions having same magnetic moments are:
   (A) $[Ti(H_2O)_6]^{3+}$ (B) $[Sc(H_2O)_6]^{3+}$
   (C) VO^{2+} (D) $[Mn(H_2O)_6]^{+3}$

7. Which of the following reactions is feasible for the preparation of 1-propoxy-2-methyl propane.
   (A) Williamson's synthesis of (Me_2CHCH_2ONa + $MeCH_2CH_2Cl$)
   (B) Williamson's synthesis of (Me_2CHCH_2Cl + $MeCH_2CH_2ONa$)
   (C) Alkoxy mercuration-demercuration of ($Me_2CH = CH_2$ + $MeCH_2CH_2OH$)
   (D) Alkoxy mercuration-demercuration of propene with Me_2CHCH_2OH

8. Which of the following are correct reactions?
   (A) 

   (B) 

   (C) 

   (D) 

Rough Work

Comprehension Type

This section contains **4 groups of questions.** Each group has 2 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct.

Paragraph for Question Nos. 9 to 10

There are 4 glass bottles containing water solutions. These bottles are not labelled. Bottles 1, 2, 3 contained colourless solutions, while Bottles 4 contained a blue solution. Each Bottle is having any one solution among the following (Each bottle is having different solution):

Copper (II) sulphate
Sodium carbonates
Lead nitrate
Hydrochloric acid

By mixing samples of the contents of the bottles in pairs, following observations are made:

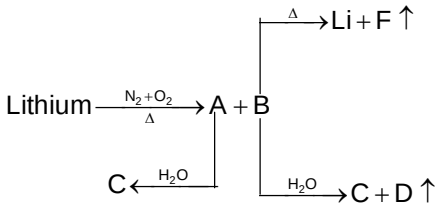
(1) Bottle 1 + Bottle 2	White ppt.
(2) Bottle 1 + Bottle 3	White ppt.
(3) Bottle 1 + Bottle 4	White ppt.
(4) Bottle 2 + Bottle 3	Colourless gas evolved
(5) Bottle 2 + Bottle 4	No visible reaction
(6) Bottle 3 + Bottle 4	Blue ppt.

9. Colourless solution present in Bottle 1 is:
   (A) $CuSO_4$ (B) $Pb(NO_3)_2$
   (C) HCl (D) Na_2CO_3

10. Chemical formula of white ppt. formed in observation (3) is:
   (A) $PbCl_2$ (B) $PbCO_3$
   (C) $CuCO_3$ (D) $PbSO_4$

Rough Work

Paragraph for Question Nos. 11 to 12



11. Which of the following is compound (A)?
(A) Li_3N (B) Li_2O
(C) Li_2O_2 (D) LiO_2

12. Which of the following is a property of gas F↑?
(A) It is paramagnetic in nature.
(B) It's bond order is 3.
(C) It does not contain lone pairs.
(D) It contains two sigma and one pi bond per molecule

Paragraph for Question Nos. 13 to 14

The contributions of both heat (enthalpy) and randomness (entropy) shall be considered to the overall spontaneity of a process. When deciding about the spontaneity of a chemical reaction or other process, we define a quantity called the Gibb's energy.

$$\Delta G = \Delta H - T\Delta S$$

Where ΔH = Enthalpy change, ΔS = entropy change.

If $\Delta G < 0$, process is spontaneous; $\Delta G = 0$, process is at equilibrium, $\Delta G > 0$, process is non-spontaneous.

13. For the change $H_2O(s)(273\ K, 2\ atm)$

$$\downarrow$$

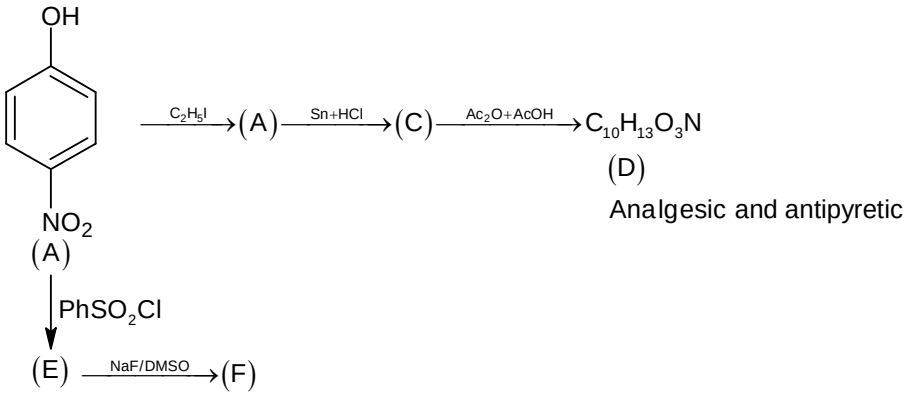
$$H_2O(\ell)(273\ K, 2\ atm)$$

Choose the correct statement:
(A) $\Delta G = 0$ (B) $\Delta G < 0$
(C) $\Delta G > 0$ (D) None of the above

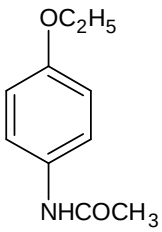
14. 5 mole of liquid water is compressed from 1 bar to 10 bar at constant temperature, change in Gibb's energy:
(ΔG) in joule is (density of water = 1000 kg/m^3)
(A) 18 (B) 225
(C) 450 (D) 900

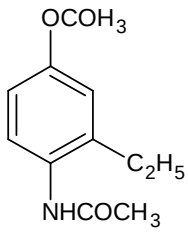
Rough Work

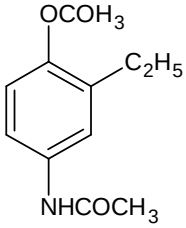
Paragraph for Question Nos. 15 to 16

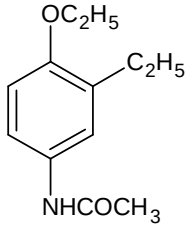


15. The compound (D) is:

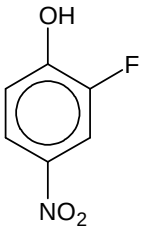
(A) 

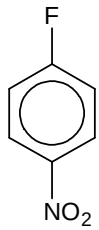
(B) 

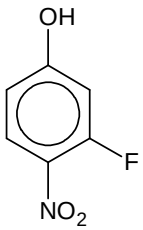
(C) 

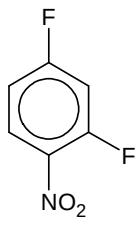
(D) 

16. The compound (F) is:

(A) 

(B) 

(C) 

(D) 

Rough Work

(Matching List Type)

This section contains **4 multiple choice questions. Each question has matching lists.** The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

17. Match List I with List II and select the correct options given below:

List – I		List – II	
(P)	For spontaneous reaction	(1)	$\sum(BE)_R - \sum(BE)_P$
(Q)	For endothermic reaction	(2)	$\Delta H = \Delta E$
(R)	Bond dissociation energy	(3)	$\Delta G = -ve$
(S)	For solids and liquids in liquids in thermo-chemical reaction	(4)	$\sum H_P > \sum H_R$

Codes:

	P	Q	R	S
(A)	3	1	2	4
(B)	2	1	4	3
(C)	3	4	1	2
(D)	4	1	2	3

18. Match List I with List II and select the correct options given below:

List – I		List – II	
(P)	10 ml 0.1 M NaOH aq. solution is added to 10 0.1 M HCl aqueous soluntion	(1)	Salt of weak acid and strong base
(Q)	10 ml, 0.1 M NaOH aq. solution is added to 10 ml 0.1 M CH_3COOH aq. solution	(2)	Salt of weak base and strong acid
(R)	10 ml 0.1 M HCl aq. solution is added to 10 ml 0.1 M NH_3 aq. solution	(3)	pH of the solution will be 7
(S)	10 ml, 0.2 M HCl aq. solution to 10 ml, 0.1 M KOH aq. solution	(4)	pH of the solution will be 1.33

Codes:

	P	Q	R	S
(A)	3	1	2	4
(B)	2	1	4	3
(C)	4	1	3	2
(D)	4	1	2	3

Rough Work

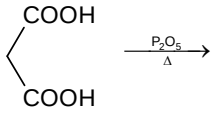
19. Match List I with List II and select the correct options given below:

List – I		List – II	
(P)	$B + N_2 \xrightarrow{\Delta}$	(1)	Inorganic graphite
(Q)	$Na_2B_4O_7(aq)$	(2)	Amphoteric
(R)	Al_2O_3	(3)	Basic
(S)	BBr_3	(4)	Lewis acid

Codes:

	P	Q	R	S
(A)	3	1	2	4
(B)	2	1	4	3
(C)	4	1	3	2
(D)	1	3	2	4

20. Match List I with List II and select the correct options given below:

List – I		List – II	
(P)	$(COOH)_2.2H_2O \xrightarrow[150^oC]{\Delta}$ (I)	(1)	$2CO_2 + H_2O$
(Q)	(I) $\xrightarrow[\text{or } H_2SO_4, \text{ at } 90^oC]{\Delta a + 200^oC}$	(2)	$(COOH)_2$
(R)		(3)	$CO + CO_2 + H_2O$
(S)	(I) $\xrightarrow[KMnO_4]{\Delta}$	(4)	Carbon sulphoxide O = C = C = C = O

Codes:

	P	Q	R	S
(A)	3	1	2	4
(B)	2	1	4	3
(C)	2	3	4	1
(D)	4	1	2	3

Rough Work

Mathematics

PART – III

SECTION – A
Straight Objective Type

This section contains **8 multiple choice questions** numbered 1 to 8. Each question has 4 choices (A), (B), (C) and (D), out of which only **ONE OR MORE** is/are correct

1. $\int e^{x^4}\left(x + x^3 + 2x^5\right)e^{x^2}dx$ is equal to

   (A) $\frac{1}{2}xe^{x^2}e^{x^4} + c$ (B) $\frac{1}{2}x^2e^{x^4} + c$

   (C) $\frac{1}{2}e^{x^2}e^{x^4} + c$ (D) $\frac{1}{2}x^2e^{x^2}e^{x^4} + c$

2. In triangle ABC, $\frac{a}{b} = \frac{2}{3}$ and $\sec^2 A = \frac{8}{5}$. Then the number of triangles satisfying these conditions is

   (A) 0 (B) 1

   (C) 2 (D) 3

3. If a team scores 200 runs for the loss of two wickets, the number of ways of distribution of runs in partnerships if there was a partnership of at least 50 runs for each wicket, is

   (A) $^{52}C_2$ (B) $^{200}C_3$

   (C) $^{50}C_2$ (D) $^{49}C_2$

4. If $\tan\theta_1$, $\tan\theta_2$, $\tan\theta_3$ are the real roots of the equation $x^3 - (a + 1)x^2 + (b - a)x - b = 0$, where $\theta_1 + \theta_2 + \theta_3 \in (0, \pi)$, then $\theta_1 + \theta_2 + \theta_3$ is equal to

   (A) $\frac{\pi}{2}$ (B) $\frac{\pi}{4}$

   (C) $\frac{3\pi}{4}$ (D) π

Space for rough work

5. If a, b, and c are in G.P. and x and y, respectively, be arithmetic means between a, b and b, c then

(A) $\frac{a}{x}+\frac{c}{y}=2$ (B) $\frac{a}{x}+\frac{c}{y}=\frac{c}{a}$

(C) $\frac{1}{x}+\frac{1}{y}=\frac{2}{b}$ (D) $\frac{1}{x}+\frac{1}{y}=\frac{2}{ac}$

6. A rod of length 2 units whose one end is (1, 0, –1) and other end touches the plane $x - 2y + 2z + 4 = 0$, then

(A) the rod sweeps the figure whose volume is π cubic units

(B) the area of the region which the rod traces on the plane is 2π

(C) the length of projection of the rod on the plane is $\sqrt{3}$ units

(D) the centre of the region which the rod traces on the plane is $\left(\frac{2}{3}, \frac{2}{3}, \frac{-5}{3}\right)$

7. $f(x)=\frac{\sqrt{x-2\sqrt{x-1}}}{\sqrt{x-1}-1}$ x, then

(A) f′(10) = 1 (B) $f'\left(\frac{3}{2}\right)=-1$

(C) domain of f(x) is $x \geq 1$ (D) range of f(x) is $(-2, -1] \cup (2, \infty)$

8. A point $P\left(\sqrt{3}, 1\right)$ moves on the circle $x^2 + y^2 = 4$ and after covering a quarter of the circle leaves it tangentially. The equation of line along which the point moves after leaving the circle is

(A) $y=\sqrt{3}x+4$ (B) $\sqrt{3}y=x+4$

(C) $y=\sqrt{3}x-4$ (D) $\sqrt{3}y=x-4$

Space for rough work

Comprehension Type

This section contains **4 groups of questions.** Each group has 2 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct.

Paragraph for Question Nos. 9 to 10

Read the following write up carefully and answer the following questions:
Tangent to the parabola $y = x^2 + ax + 1$, at the point of intersection of y–axis also touches the circle $x^2 + y^2 = r^2$. Also no point of the parabola is below x–axis

9. The radius of circle when a attains its maximum value
   (A) $\frac{1}{\sqrt{10}}$ (B) $\frac{1}{\sqrt{5}}$
   (C) 1 (D) $\sqrt{5}$

10. The slope of the tangent when radius of the circle is maximum is
   (A) –1 (B) 1
   (C) 0 (D) 2

Paragraph for Question Nos. 11 to 12

Read the following write up carefully and answer the following questions:
Consider the equation $az + b\bar{z} + c = 0$, where a, b, c $\in$ **C** (set of complex number)

11. If |a| = |b| and $\bar{a}c \neq b\bar{c}$, then z has
   (A) infinite solutions (B) no solutions
   (C) finite solutions (D) cannot say anything

12. If |a| = |b| $\neq 0$ and $\bar{a}c = b\bar{c}$, then $az + b\bar{z} + c = 0$ represents
   (A) an ellipse (B) a circle
   (C) a point (D) a straight line

Space for rough work

Paragraph for Question Nos. 13 to 14

Read the following write up carefully and answer the following questions:

If e^A is defined as $e^A = I + A + \frac{A^2}{2!} + \frac{A^3}{3!} + ... = \frac{1}{2}\begin{bmatrix} f(x) & g(x) \\ g(x) & f(x) \end{bmatrix}$ where $A = \begin{bmatrix} x & x \\ x & x \end{bmatrix}$ and $0 < x < 1$, I is an identify matrix

13. $\int \frac{g(x)}{f(x)} dx$ is equal to

(A) $\log(e^x + e^{-x}) + c$ (B) $\log(e^x - e^{-x}) + c$

(C) $\log(e^{2x} - 1) + c$ (D) none of these

14. $\int (g(x)+1)\sin x dx$ is equal to

(A) $\frac{e^x}{2}(\sin x - \cos x) + c$ (B) $\frac{e^{2x}}{5}(2\sin x - \cos x) + c$

(C) $\frac{e^x}{5}(\sin 2x - \cos 2x) + c$ (D) none of these

Paragraph for Question Nos. 15 to 16

Read the following write up carefully and answer the following questions:

f(x) satisfies the relation $f(x) - \lambda \int_0^{\pi/2} \sin x \cos t\, f(t) dt = \sin x$

15. If f(x) = 2 has at least one real root, then

(A) $\lambda \in [1, 4]$ (B) $\lambda \in [-1, 2]$

(C) $\lambda \in [0, 1]$ (D) $\lambda \in [1, 3]$

16. If $\int_0^{\pi/2} f(x) dx = 3$, then the value of λ is

(A) 1 (B) $\frac{3}{2}$

(C) $\frac{4}{3}$ (D) none of these

Space for rough work

(Matching list Type)

This section contains **4 multiple choice questions.** Each question has matching lists. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

17. Match the following List–I with List–II.

	List – I		List – II
(P)	Four dice (six faced) are rolled. The number of possible outcomes in which at least one dice shows 2 is	1.	210
(Q)	Let A be the set of 4–digit number $a_1a_2a_3a_4$ where $a_1 > a_2 > a_3 > a_4$. Then n(A) is equal to	2.	750
(R)	The total number of three–digit numbers, the sum of whose digits is even, is equal to	3.	671
(S)	The number of four–digit numbers without repetition that can be formed from the digits 0, 1, 2, 3, 4, 5, 6, 7 so that each number contains digit 1 is	4.	450

Codes:

	P	Q	R	S
(A)	4	2	1	3
(B)	3	1	4	2
(C)	1	3	2	4
(D)	3	4	1	2

18. Match the following List–I with List–II.

	List – I		List – II
(P)	Diameter of the circle touching the line $(x - 1)\cos\theta + (y - 1)\sin\theta = 1$ for all values of θ	1.	2
(Q)	Radius of smallest circle which touches the circle $x^2 + y^2 = 1$ and $x^2 + y^2 - 6x - 8y + 21 = 0$	2.	1
(R)	Number of values of m for which the line $(y - 2) = m(x - 1)$ cuts the circle $x^2 + y^2 = 5$ at two real points	3.	4
(S)	Number of circle touching both the axes and the line $x + y = 4$	4.	infinite

Codes:

	P	Q	R	S
(A)	1	2	3	4
(B)	3	4	1	2
(C)	1	2	4	3
(D)	2	4	3	1

Space for rough work

19. Match the following List–I with List–II.

	List – I		List – II
(P)	Least value of the expression $2\log_{10} x - \log_x 0.01$ for $x > 1$ is	1.	2
(Q)	The number of solutions of $\log_4(x-1) = \log_2(x-3)$ is	2.	4
(R)	Least integral value of x which satisfies $\log_{x+1}\sqrt{2}x > \frac{1}{2}$ is	3.	1
(S)	Number of solutions of $[\log_e x] = \lvert x \rvert$ is (where [.] denotes the greatest integer function)	4.	0

Codes:

	P	Q	R	S
(A)	2	3	4	1
(B)	2	1	2	4
(C)	2	4	3	1
(D)	2	3	1	4

20. Match the following List–I with List–II.

	List – I		List – II
(P)	If f : (0, ∞) onto (0, ∞) be a derivable function for which there exists its primitive F such that $2(F(x) - f(x)) = f^2(x)$ for any real positive x, then $\lim_{x\to\infty} \frac{f(x)}{x}$ is equal to	1.	$\frac{1}{3}$
(Q)	If f(x) is continuous and differentiable over [–2, 5] and $3 \le f'(x) \le 4$ for all x in (–2, 5), then value of f(5) – f(–2) may be	2.	1
(R)	If x is real, then the greatest value of the expression $\frac{x+2}{2x^2+3x+6}$ is	3.	2
(S)	If $y = e^{-x}\cos x$ and $y_4 + 2ky = 0$, where $y_4 = \frac{d^4y}{dx^4}$, then k is equal to	4.	22

Codes:

	P	Q	R	S
(A)	4	1	2	3
(B)	3	1	4	2
(C)	2	4	1	3
(D)	2	4	3	1

Space for rough work